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import cv2
import numpy as np
from google.colab.patches import cv2_imshow

# Read the image
img = cv2.imread("lab_image.jpg")

# Check if image is loaded
if img is None:
    print("Image not found!")
else:
    # Laplacian mask with diagonal neighbors
    kernel = np.array([[1, 1, 1],
                       [1, -8, 1],
                       [1, 1, 1]])

    # Apply sharpening filter
    sharpened = cv2.filter2D(img, -1, kernel)

    # Save output image
    cv2.imwrite("Sharpened_Image.jpg", sharpened)

    # Display images
    print("Original Image")
    cv2_imshow(img)

    print("Sharpened Image")
    cv2_imshow(sharpened)
```

Original Image



Sharpened Image



